

Analysis for Similar Ventures - Maroon Cays Pitch

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There are currently no ventures in The Bahamas that integrate a dedicated research center combining solar energy innovation, waste-to-energy systems, and on-site accommodation for researchers within a single operational platform. However, as of December 2025, the Davis administration has advanced a large-scale renewable energy venture through a \$40 million hybrid energy project across Cat Island, Long Island, and San Salvador, with completion expected in 2027. Their project is expected to combine solar PV, battery storage, and natural gas generation to meet 80–90% of each island's energy demand. Nonetheless, the project by the Davis administration doesn't compare to the Research center in our proposal since it does not provide room for testing research, conferences and accommodation for researchers.

Similarly, regional efforts like the Grenada Waste-to-Energy Recovery solely focuses on waste to energy production. Therefore this effort is also designed primarily for power generation and operational reliability rather than applied research, technology validation, or commercialization. This gap presents a strategic opportunity: while existing ventures strengthen national energy supply, they do not provide the applied research and innovation ecosystem necessary to de-risk emerging technologies, generate performance data under extreme climate conditions, or support scalable replication across Small Island Developing States.

To fill this gap, Maroon Cays is positioned to move beyond energy generation and establish itself as the infrastructure that enables innovation, validation, and commercialization within island environments. The island's long, narrow topography, proximity to George Town airport, and existing off-grid infrastructure create a controlled but realistic testing environment. Technologies can be evaluated for hurricane resilience, salt-air corrosion resistance, humidity durability, grid integration, and long-term degradation performance. On-site data acquisition systems, IV curve tracing, thermal imaging, battery cycling labs, and monitored solar tracker fields will generate performance data that is currently unavailable in the Bahamian market. This transforms Maroon Cays from a development project into a data-producing infrastructure asset.

Strategically, Maroon Cays fills a structural gap between laboratory research institutions such as the National Renewable Energy Laboratory and Caribbean energy deployment projects. Mainland labs cannot replicate island climate stressors, while existing Bahamian energy projects do not offer controlled testing or commercialization support. Maroon Cays bridges this divide by enabling technology validation at the point of intended distribution across Small Island Developing States. Therefore, Maroon Cay is positioning itself as a climate infrastructure proving ground for coastal and island energy systems. By integrating research facilities, short-term researcher accommodations, and industry conference capabilities into a single operational campus, the center creates recurring partnership opportunities with universities, clean energy firms, and modular waste innovators. Its value proposition is clear: test where you deploy, validate before you scale, and generate location-specific performance data that accelerates commercialization across island markets.

Citation

Grenada Solid Waste Management Authority. (2024, February 19). Grenada Solid Waste Management Authority. <https://www.gswma.com/>

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